Power Stage

File: power_stage.kicad_sch

Primary Gate Drive

File: gate_drive.kicad_sch

Control and SR Drive

File: control.kicad_sch

SiC ZVS PHASE-SHIFTED FULL-BRIDGE ISOLATED DC-DC CONVERTER - 3 kW

Input 370-420 V DC (400 V nominal, PFC output) | Output 48 V / 62.5 A | 100 kHz | reinforced isolation 4 kVrms

Primary: 4 x C3M0060065K SiC MOSFET full bridge, 12 uH resonant inductor, C3D04065A clamp diodes, 4.7 uF DC-blocking

Transformer: ETD59 3C95, 18:3:3 centre-tapped. Secondary: 4 x IPP110N20N3 synchronous rectifiers, 6.8 uH / 2.8 mF output filter

Control: UCC28951 (secondary side, peak current mode) -> UCC21520 isolated drivers (primary), UCC27524 (SR)

Gate bias: 4 x R12P21503D isolated +15 V / -3 V modules. Bias input: +12 V aux (secondary referenced).

Sheets: 2 - Power Stage, 3 - Primary Gate Drive, 4 - Control and SR Drive. Generated from tools/kicad (python).

WARNING: primary side nets (VBUS, PGND, SW_A, SW_B, LR_N, PRI_x, G_Q1..4, KS_Q1..4, VDD/VEE_Q1..4) are at hazardous voltage.

3 kW, 400 V -> 48 V, 100 kHz, ZVS phase-shifted full bridge

Root sheet

Open hardware - SiC PSFB reference design

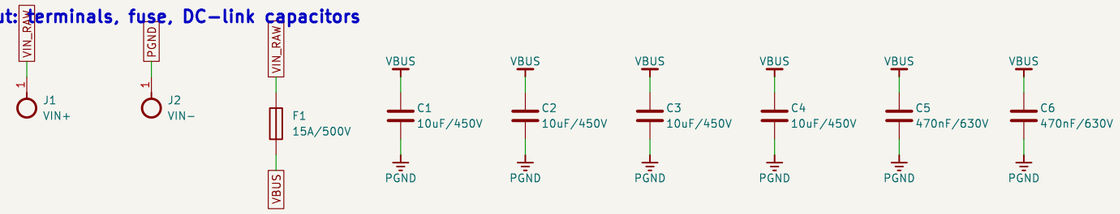
Sheet: /

File: SiC_PSFB_3kW.kicad_sch

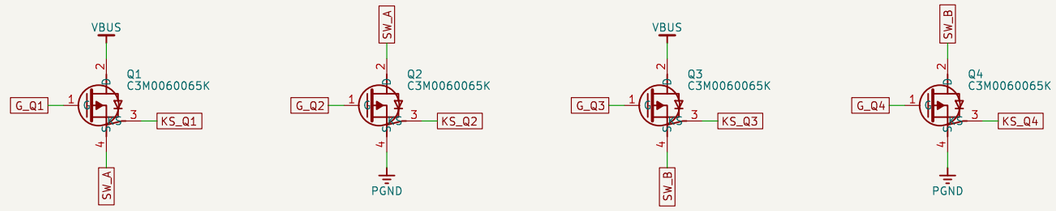
Title: SiC ZVS Phase-Shifted Full-Bridge 3 kW DC-DC

Size: A3	Date: 2026-10-09	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/4

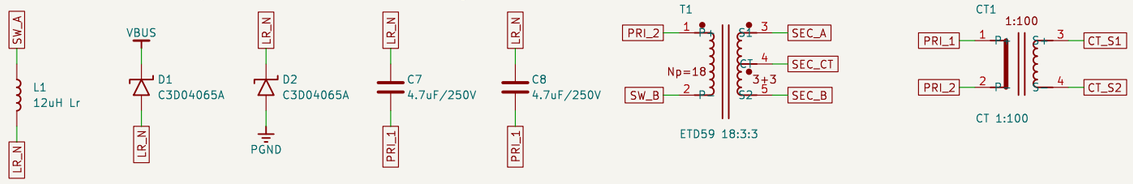
Input: terminals, fuse, DC-link capacitors



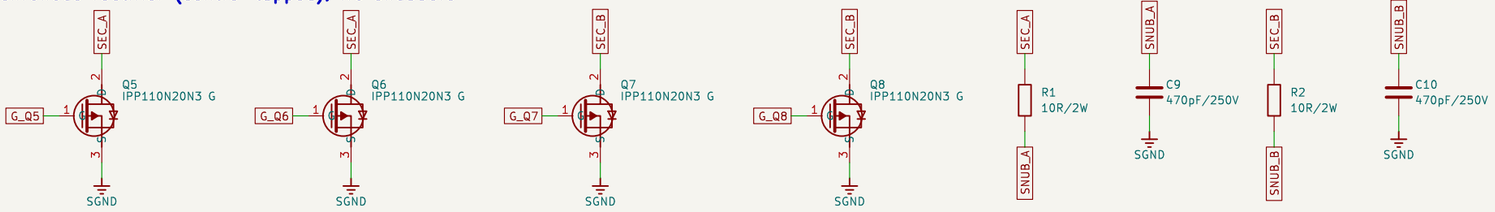
SiC full bridge Q1–Q4 (leg A = Q1/Q2 driven by OUTA/OUTB, leg B = Q3/Q4 by OUTC/OUTD)



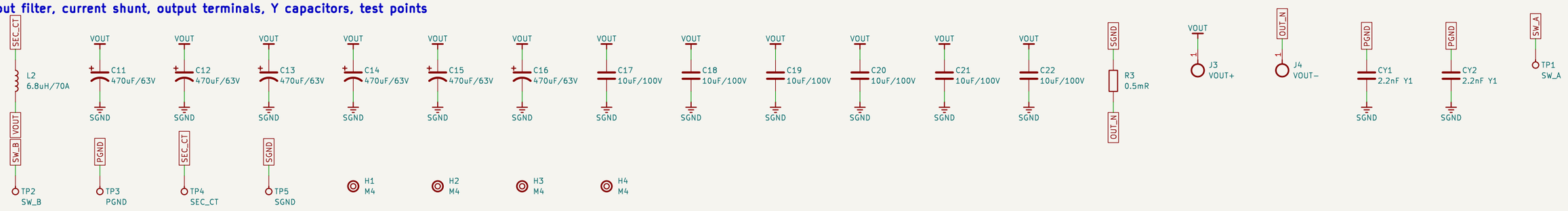
Resonant inductor, primary clamp diodes, DC blocking, main transformer, current transformer



Synchronous rectifier (centre-tapped), RC snubbers



Output filter, current shunt, output terminals, Y capacitors, test points



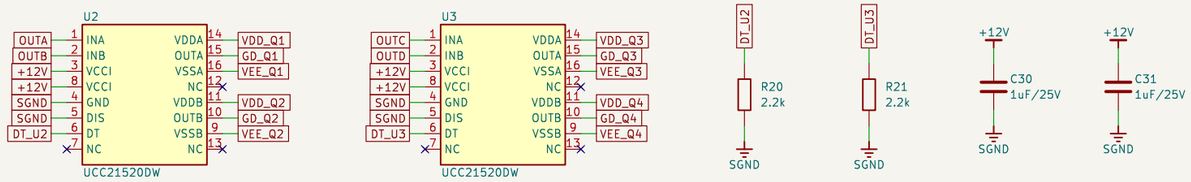
3 kW, 400 V –> 48 V, 100 kHz, ZVS phase-shifted full bridge
Power Stage
Open hardware — SiC PSFB reference design

Sheet: /Power Stage/
File: power_stage.kicad_sch

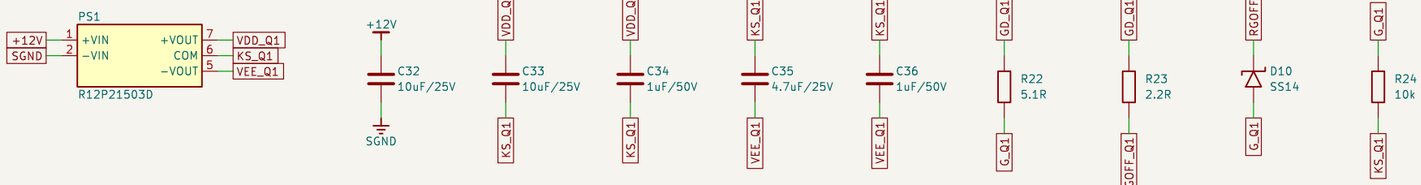
Title: SiC ZVS Phase-Shifted Full-Bridge 3 kW DC-DC

Size: A3	Date: 2026-10-09	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

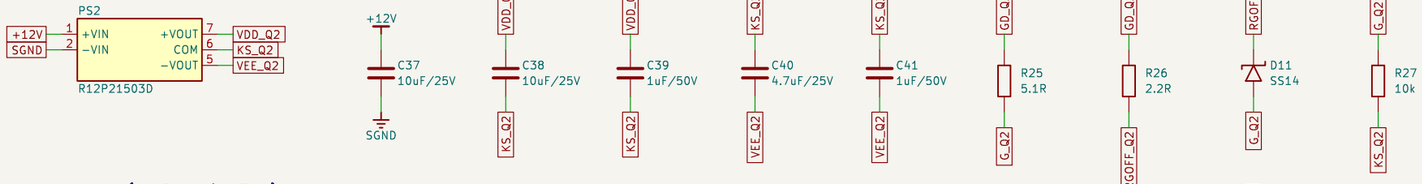
Isolated gate drivers UCC21520 (logic side = secondary ground, driver side = per-MOSFET Kelvin source)



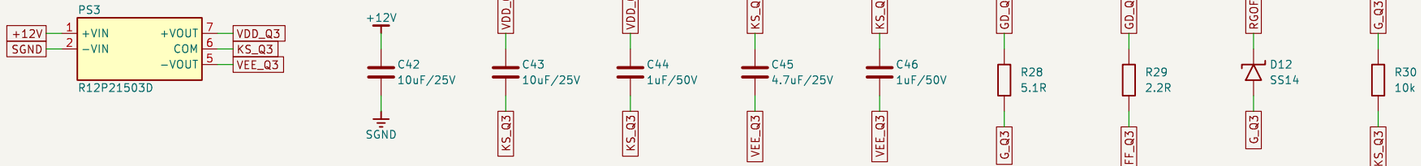
Q1 isolated bias (+15 V / −3 V) and gate network



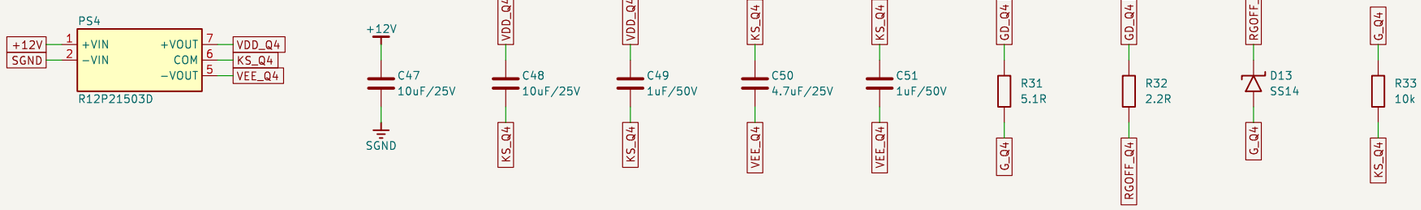
Q2 isolated bias (+15 V / −3 V) and gate network



Q3 isolated bias (+15 V / −3 V) and gate network



Q4 isolated bias (+15 V / −3 V) and gate network



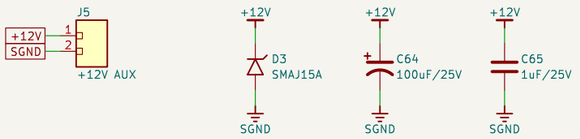
3 kW, 400 V -> 48 V, 100 kHz, ZVS phase-shifted full bridge
Primary Gate Drive
Open hardware — SiC PSFB reference design

Sheet: /Primary Gate Drive/
File: gate_drive.kicad_sch

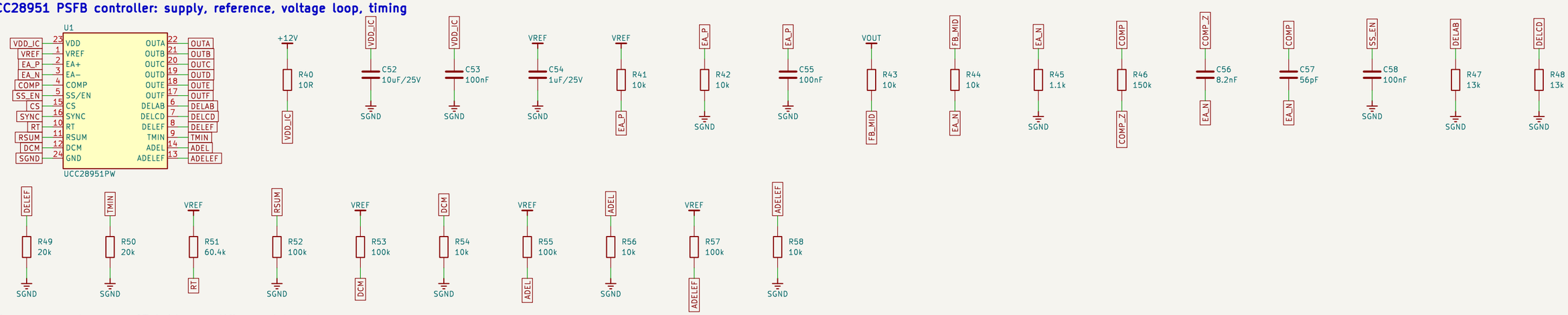
Title: SiC ZVS Phase-Shifted Full-Bridge 3 kW DC-DC

Size: A3	Date: 2026-10-09	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/4

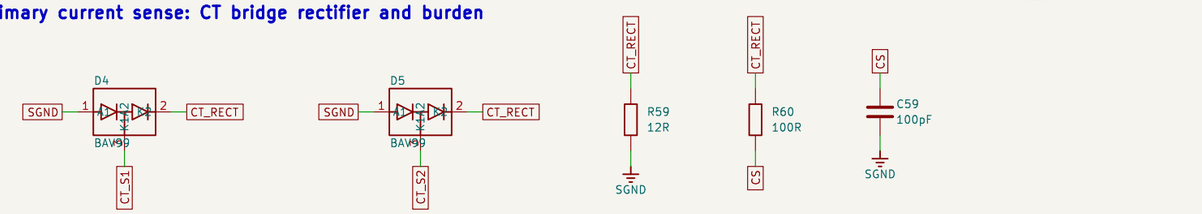
Auxiliary 12 V bias input



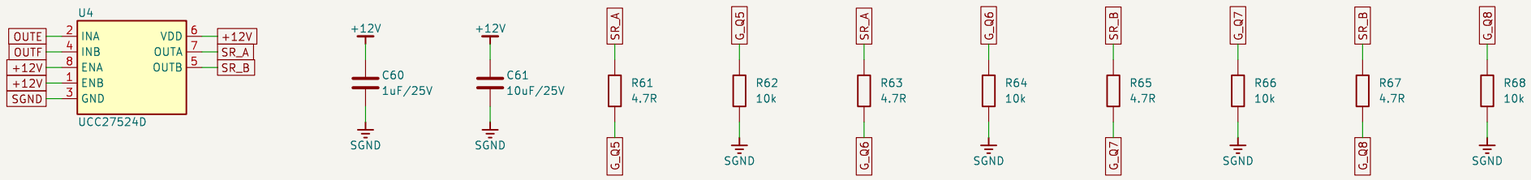
UCC28951 PSFB controller: supply, reference, voltage loop, timing



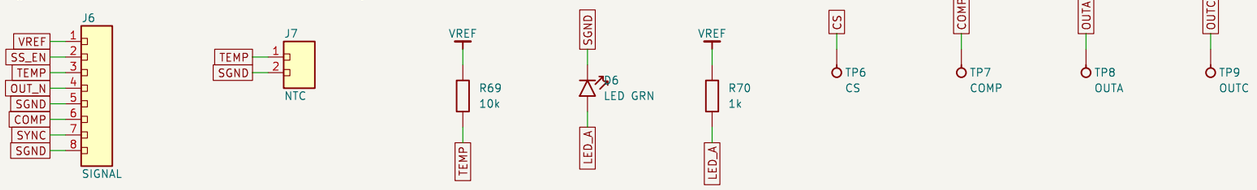
Primary current sense: CT bridge rectifier and burden



Synchronous rectifier gate drive UCC27524



Signal connector, NTC, indicator, test points



3 kW, 400 V –> 48 V, 100 kHz, ZVS phase–shifted full bridge
Control and SR Drive
Open hardware – SiC PSFB reference design

Sheet: /Control and SR Drive/
File: control.kicad_sch

Title: SiC ZVS Phase–Shifted Full–Bridge 3 kW DC–DC

Size: A3	Date: 2026–10–09	**Rev: A**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 4/4